Microsoft

TD SYNNEX

GitHub

Technical Walkthrough

Russ Seaman
Solutions Architect II · Microsoft Cloud, Power Platform & GitHub
russ.seaman@tdsynnex.com · linkedin.com/in/russseaman

October 2026 · TD SYNNEX Cloud Enablement Services

TD SYNNEX

Agenda

A practical introduction to GitHub: how git thinks, how Enterprise is put together, and the workflows, automation, and practices that make teams productive — two hours, one break, three live demos.

1

8 MIN

Git vs. GitHub

What each one is, and why Enterprise exists

2

25 MIN

How git works

Snapshots, branches, merges — and version history on a real deck

3

21 MIN

Inside Enterprise

Orgs, identity, policies — and a dev environment in one click

4

15 MIN

Best practices

Repo anatomy, branching, pull requests, rulesets

5

30 MIN

Automate & secure

Actions, Copilot, security — then Copilot builds an app, safely

6

6 MIN

Admin & buying

What it costs, and how partners buy

TD SYNNEX

01

Git vs. GitHub

PART 1 — What each one is, and why Enterprise exists · 8 min

Git is the engine; GitHub is everything built around it

TD SYNNEX

THE TOOL

git

"the stupid content tracker" — its own manual page

- Version control that runs on your laptop — no server needed
- Created in 2005 by Linus Torvalds to manage the Linux kernel
- Free, open source, and everywhere: Windows, Mac, Linux, Codespaces
- Knows nothing about pull requests, issues, or permissions

push





pull

THE PLATFORM

GitHub

The place teams meet around the code

- Hosts your git repositories in the cloud — the "remote"
- Adds collaboration: pull requests, code review, issues, projects
- Adds automation and safety: Actions, security scanning, Copilot
- Enterprise adds identity, policy, and audit on top

SAME GIT, ANY HOST

GitHub, GitLab, Azure Repos, Bitbucket and a bare server all speak the same git. The commands you learn in Part 2 work everywhere; what differs is everything on the right.

▸ **You'll use git every hour and GitHub every day — learn the tool first, then the platform.**

Enterprise adds the control layer

TD SYNNEX

Same platform at every tier — what changes is how much control IT has over it.

Free

$0

- Unlimited public & private repos
- 2,000 Actions minutes / month
- 500 MB Packages storage
- Community support

Team

$4 / user / month *

- Everything in Free, plus:
- 3,000 Actions minutes / month
- 2 GB Packages storage
- Codespaces for the org
- Protected branches, required reviewers, code owners on private repos

Enterprise

$21 / user / month *

- Everything in Team, plus:
- 50,000 Actions minutes · 50 GB Packages
- Enterprise Managed Users + SAML SSO
- Enterprise account: policies across every org
- Audit log API & streaming, IP allow list
- Internal repos, enterprise rulesets, data residency
- Cloud or self-hosted Server — same license

* github.com/pricing, Aug 2026 — Team and Enterprise are listed "per user/month for the first 12 months"; renewal pricing is not published — confirm it with GitHub or your Microsoft rep.

- **Team is for a team; Enterprise is for a company — identity, policy, and audit are what you pay for.**

TD SYNNEX

02

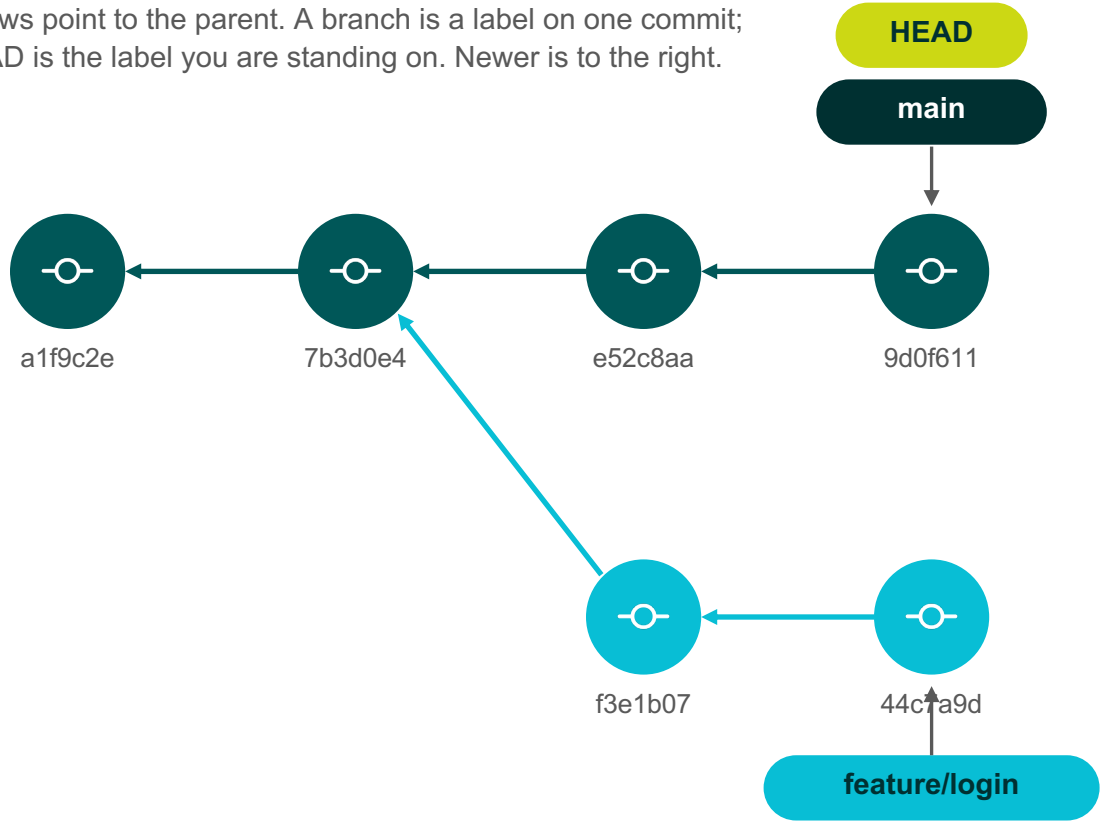
How git works

PART 2 — Snapshots, branches, merges — then version history on a real deck · 25 min

A commit is a snapshot, not a diff

TD SYNNEX

Arrows point to the parent. A branch is a label on one commit; HEAD is the label you are standing on. Newer is to the right.



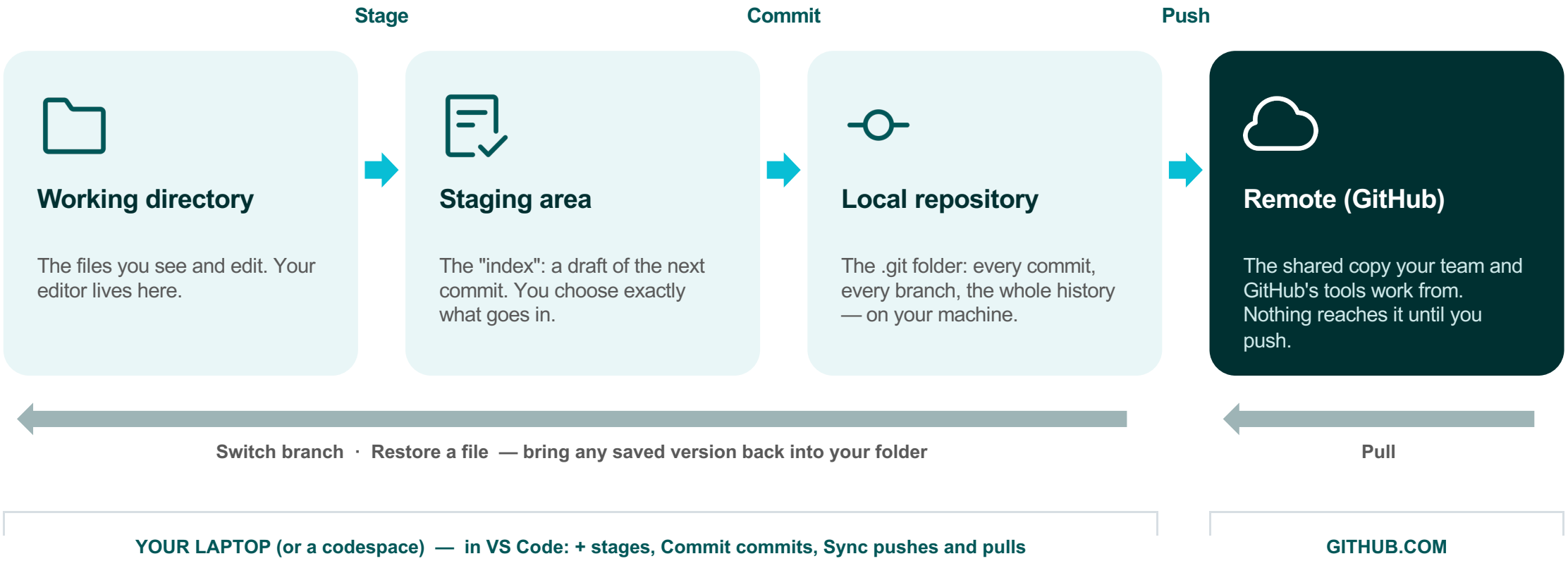
INSIDE ONE COMMIT

- **A snapshot of every file**
  a "tree" — not a list of changes; unchanged files are shared, so it stays small
- **The parent commit ID**
  two parents for a merge — this is what makes it a chain
- **Author, date, message**
  who, when, and why — the message is for humans
- **An ID computed from all of it**
  the SHA (e.g. 9d0f611…). Change one byte anywhere and every later ID changes — history is tamper-evident

- **Because every commit points to its parent, git can rebuild any version — and nothing is rewritten silently.**

Every change moves through four places

TD SYNNEX



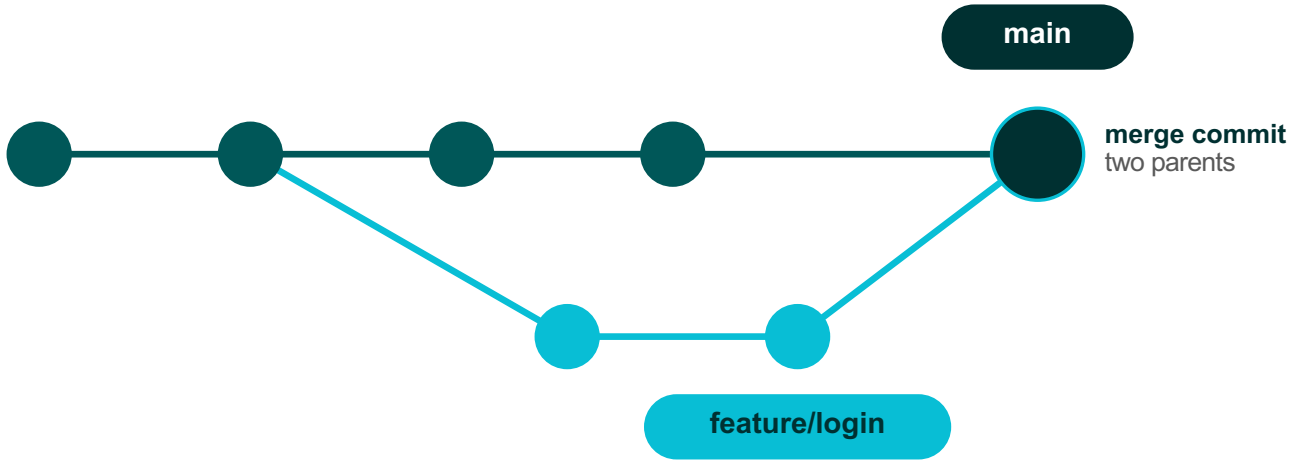
- **Stage, commit, push — say those three verbs and you understand git.**

Making IT Personal™

Branches are pointers, so they're cheap — and a conflict is just a choice

TD SYNNEX

A branch is a pointer to a commit; creating one costs nothing. Merging brings its commits back to main.



FAST-FORWARD

main has not moved since you branched, so the pointer just slides forward. No merge commit.

MERGE COMMIT

Both sides moved: one new commit with two parents ties them together. History stays honest.

REBASE

Replay your commits on top of the latest main for a straight line. Never rebase a branch others share.

WHEN TWO PEOPLE EDIT THE SAME LINE

```
<<<<<<< HEAD
Welcome to Contoso Bikes
=======
Welcome to Contoso Cycles
>>>>>>> feature/rename
```

1. Open the file; keep one side (or both)
2. Delete the <<< === >>> marker lines
3. Stage the file, then commit

- **Branch for every change, merge through a pull request — and on a conflict, you decide, not git.**

Making IT Personal™

Demo 1 · Version history you control

TD SYNNEX

LIVE DEMO · 12 MIN — Just watch

1 Create the shared copy

github.com → New repository → contoso-proposal, private, with a README.

2 Clone it in VS Code

Source Control → Clone Repository → Clone from GitHub. GitHub has the shared copy; this one is mine.

3 Save version 1

Drop in the deck and the notes. Stage (+), say why, Commit, Sync. Refresh GitHub: they're there.

4 Change and compare

Retitle a slide, edit a line of notes. The notes show red and green; the deck just says binary. Commit.

5 The time machine

Delete the pricing slide and sync. Then pull the old version from GitHub's history and bring it back.

WATCH FOR ✓ **the M after every save** ✓ **red and green for text, binary for the deck** ✓ **every version on GitHub**

IF IT BREAKS the first Sync asks you to sign in: approve it in the browser. If .DS_Store shows up, right-click it → Add to .gitignore.

TD SYNNEX

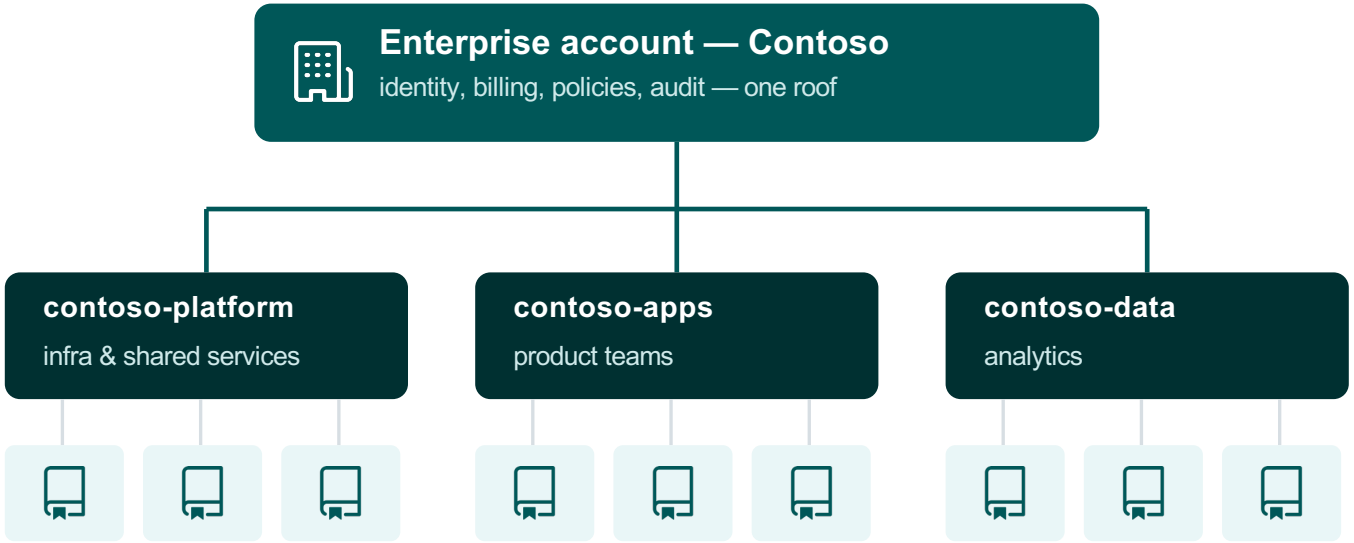
03

Inside GitHub Enterprise

PART 3 — Enterprise → organizations → repositories; roles, identity, and policies · 21 min

One enterprise, many orgs, thousands of repos

TD SYNNEX



repositories — the code, issues, PRs, and settings, always owned by an organization

Teams group people inside an org and carry repo permissions. Sync them from Entra ID groups so joiners and leavers are handled by HR, not by an admin.

THREE REPO VISIBILITIES

Public

Anyone on the internet. Managed-user enterprises cannot create these.

Internal

Everyone in the enterprise, nobody outside — "innersource". Enterprise plan only.

Private

Only the people and teams you grant.

- **Policy and identity live at the enterprise; people and permissions live in orgs; code lives in repos.**

Six roles — only two of them are admins

TD SYNNEX

ROLE	LIVES AT	WHAT THEY CAN DO	USES A LICENSE?
Enterprise owner	Enterprise	Everything: policies, billing, licenses, every org, audit log. Have at least two.	ONLY IF IN AN ORG
Organization owner	Organization	Members, teams, repos, settings, security and Copilot for that org.	YES
Member	Organization	Works in repos through team permissions and the org base permission.	YES
Outside collaborator	Repository	Access to specific repos without joining the org — vendors, contractors.	PRIVATE/INTERNAL: YES
Guest collaborator	Enterprise (EMU)	A managed user limited to the repos they are added to — internal repos stay hidden.	NOT UNTIL IN AN ORG
Billing manager	Enterprise / org	Sees and manages billing only — ideal for a partner staying attached.	NO

- **Give the fewest people the most power: two-plus enterprise owners, an owner per org, everyone else in a team.**

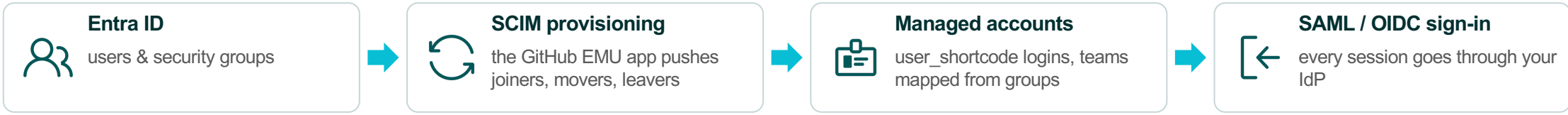
With EMU your IdP owns the accounts — otherwise your users do

TD SYNNEX

Enterprise Managed Users (EMU)	Personal accounts + SAML SSO
◆ Accounts are created, updated and removed by your IdP over SCIM — Entra ID, Okta, PingFederate	◆ Developers keep their own github.com identity; SSO links it to your enterprise
◆ Sign-in only through your IdP (SAML or OIDC); usernames carry your enterprise shortcode	◆ Open-source and cross-company collaboration stays natural
◆ No public repos or gists, no collaboration outside the enterprise, no personal use	◆ You govern access to your resources, not the account itself
◆ Required for GHE.com data residency	◆ SCIM provisioning and team sync work per organization; enterprise-level SAML has no SCIM

PICK EMU WHEN the customer is regulated, Entra-centric, and does not need staff contributing to open source from work accounts. Pick personal accounts + SSO when developers live in open source or work across companies.

HOW EMU PLUGS INTO ENTRA ID



- **Decide identity before the first repo is created — moving between the two models later means a migration.**

Making IT Personal™

Cloud first; Server when you must

Same per-user license, three ways to run it:

1

Enterprise Cloud

github.com, run by GitHub. Every feature and Copilot capability lands here first; EMU or personal accounts; nothing to patch. Pick it unless a rule says otherwise.

2

Data residency

Enterprise Cloud on your own GHE.com subdomain — contoso.ghe.com — with data stored in the EU, US, Australia, or Japan. Requires Enterprise Managed Users; no public repos; some Copilot features lag.

3

Enterprise Server

Self-hosted on your VMs or cloud; you run upgrades, backups and scale. Version 3.21 (June 2026). Same license; GitHub Connect bridges to the cloud for Copilot and Actions.

Demo 2 · A dev environment in one click

TD SYNNEX

LIVE DEMO · 8 MIN — Just watch

1 One click

Code → Codespaces → Create. Before it builds, GitHub shows who pays: the org.

2 While it builds

Org settings → Codespaces: who can use it, machine sizes, idle timeout, and the budget.

3 Copilot and rules

Copilot seats and policies; the ruleset that sends every change to main through a pull request.

4 The audit trail

Audit log, filtered to action:codespaces: the one you just made is already on the record.

5 Ready to code

Node and Copilot are already installed. Open devcontainer.json: one file is everyone's setup.

WATCH FOR ✓ **who pays, before you click** ✓ **rules set once for the whole org** ✓ **zero installs**

IF IT BREAKS can't create a codespace? The org hasn't allowed Codespaces or set a budget. A stopped codespace restarts in one click.

TD SYNNEX

04

Structure & best practices

PART 4 — Repo anatomy, branching, pull requests, rulesets · 15 min

A well-structured repo explains itself

TD SYNNEX

```
contoso-bikes/
├─ README.md
├─ LICENSE
├─ .gitignore
├─ CONTRIBUTING.md
├─ SECURITY.md
├─ .github/
│  ├─ CODEOWNERS
│  ├─ ISSUE_TEMPLATE/
│  ├─ PULL_REQUEST_TEMPLATE.md
│  ├─ dependabot.yml
│  └─ workflows/ci.yml
├─ src/
├─ tests/
└─ docs/
```

README.md

What it is, how to run it, who owns it — the first 30 seconds of every visitor.

LICENSE

What others may do with the code. No file means all rights reserved — even in public.

.gitignore

Never commit build output, node_modules, .env files or secrets. Start from GitHub's templates.

CODEOWNERS

Maps paths to teams; owners are auto-requested as reviewers and can be required by a ruleset.

.github/

GitHub's config folder: issue and PR templates, Dependabot, and the workflows Actions runs.

SECURITY.md

How to report a vulnerability privately — shown on the Security tab.

- **If a new hire can't run it from the README in ten minutes, the repo isn't finished.**

Pick the simplest branching model that fits

TD SYNNEX

SIMPLEST

Trunk-based

Everyone integrates into main daily through tiny, short-lived branches (hours, not weeks); feature flags hide unfinished work. Needs strong automated tests. How high-velocity product teams ship.

RECOMMENDED DEFAULT

GitHub Flow

START HERE

Branch from main → commit → open a pull request → review + checks → merge → deploy. One long-lived branch, one rule: main is always deployable. Fits most teams and everything you do today.

MOST CEREMONY

Git Flow

develop, release, and hotfix branches with numbered releases. Fits packaged software that supports several versions at once; overhead for web apps and internal tools.

← FEWER BRANCHES, FASTER MERGES

MORE STRUCTURE, MORE PROCESS →

- **Start with GitHub Flow; earn your way to trunk-based; adopt Git Flow only if you ship numbered versions.**

The pull request is where quality happens

TD SYNNEX

1 Branch

feature/…, from up-to-date main. One purpose per branch.

2 Commit

Small and often; a message that says why. Push early.

3 Open the PR

Template fills in what/why/how-to-test. Link the issue. Draft while it is not ready.

4 Review

CODEOWNERS auto-requested; Copilot code review first pass; comments, suggestions, approvals.

5 Checks

Actions runs tests and scans; required checks must be green.

6 Merge

Merge commit, squash, or rebase — pick one per repo. Branch auto-deletes.

7 Deploy & close

main deploys; "Closes #231" closes the issue. Start the next branch.

↺ main stays green

Every change takes the same loop — that predictability is the point.

MERGE BUTTON **Merge commit** keeps every commit and adds one with two parents · **Squash** collapses the branch into one tidy commit (most teams' default) · **Rebase** replays commits onto main with no merge commit

- **Small PRs get reviewed in an hour; big PRs get approved unread. Keep them under a few hundred lines.**

Rulesets are guardrails, not gates

TD SYNNEX

protect-main

ACTIVE

Branch ruleset · target: main · bypass: none

- **Require a pull request before merging**
  1 approval · dismiss stale approvals · require review from code owners
- **Require status checks to pass**
  ci / test must be green; branch up to date
- **Require conversation resolution**
  every review thread resolved before merge
- **Block force pushes**
  history on main can never be rewritten
- **Restrict deletions**
  nobody deletes main by accident
- **Require signed commits**
  optional — cryptographic proof of author
- **Require linear history**
  optional — forces squash or rebase merges

EVALUATE MODE

Enterprise plans can turn a ruleset on without enforcing it: the insights page shows what would have been blocked. Roll out with zero surprises.

BYPASS LIST

Name the roles, teams or apps that may bypass — with a reason logged. Keep it empty on main if you can.

SCOPE & LAYERS

Repo, org, or enterprise level. Rulesets stack and the strictest rule wins. Push rules block large files or paths before they land.

▸ **Protect main on day one — PR required, one review, checks green, no force-push — then let evaluate mode teach you the rest.**

TD SYNNEX

05

Automate, secure, and add AI

PART 5 — Actions, Copilot, security — then Copilot builds an app, safely · 30 min

Actions runs your checks on every pull request

TD SYNNEX

.github/workflows/ci.yml

```yaml
name: ci
on:
  pull_request:
    branches: [main]
jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v7
      - uses: actions/setup-node@v7
        with:
          node-version: 24
      - run: npm ci
      - run: npm test
```



WORKFLOW

A YAML file in .github/workflows. It wakes up on events — a pull request, a push, a schedule, a button — and runs jobs.



JOBS & STEPS

Jobs run in parallel on fresh machines; steps run commands or reuse actions from the Marketplace (checkout, setup-node, deploy to Azure…).



RUNNERS

GitHub-hosted (Ubuntu, Windows, macOS) billed per minute beyond what is included, or self-hosted on your own machines for private networks and special hardware.

50,000 MIN / MO INCLUDED ON ENTERPRISE Public repos run free. Hosted-runner prices fell up to 39% in Jan 2026.

- **A required status check turns "please run the tests" into "the merge button is grey until they pass."**

Copilot is a teammate, not just autocomplete

TD SYNNEX

Completions

Suggests as you type; next-edit suggestions. Unlimited — not metered.

Chat

Ask about the code in VS Code, on github.com, on mobile, in the CLI.

Agent mode

Multi-file changes in your editor; runs the tests and fixes what fails.

Cloud agent

Assign an issue; Copilot works in its own environment and opens a PR.

Code review

Reviews every PR with suggestions you apply in one click.

Spaces & CLI

Shared context for a project; terminal help and commands.

Copilot Business — $19 / user / mo

1,900 AI credits per user per month

Any organization. Central policies, seat management, IP indemnity, audit.

Copilot Enterprise — $39 / user / mo

3,900 AI credits per user per month

Requires GitHub Enterprise Cloud. Everything in Business plus priority access to premium models and enterprise-wide context.

FOR ADMINS

Enable at the enterprise or org, assign seats (from 1 Sep 2026 self-serve card/PayPal customers pay for each new seat before access), set policies for models, agents and content exclusion, set credit budgets, watch usage. Credits pool across the enterprise; overage is $0.01 per credit unless you cap it.

Seat prices and included credits: docs.github.com Copilot plans and billing, Aug 2026. Since 1 Jun 2026 Copilot bills AI-model use in credits (1 credit = $0.01); completions stay unmetered.

▸ **Price the seat, then govern the credits — that is the new Copilot conversation.**

Secret Protection and Code Security stop the two most common breaches

TD SYNNEX

GitHub Secret Protection

$19 / active committer / mo

- Secret scanning finds leaked keys, tokens and passwords — across the whole history
- Push protection blocks a commit that contains a secret before it ever lands
- AI detection for unstructured secrets, custom patterns, validity checks, alerts
- Free on public repos; available on Team (metered) and Enterprise

GitHub Code Security

$30 / active committer / mo

- Code scanning with CodeQL finds vulnerabilities in your own code on every PR
- Copilot Autofix proposes the fix in the pull request
- Dependabot premium: dependency review, security campaigns, prioritization
- Security overview: risk and coverage across every org in the enterprise

ACTIVE COMMITTER anyone whose commit was pushed to a licensed repo in the last 90 days — counted once no matter how many repos. Buy either product alone; a combined GitHub Advanced Security license that includes both still exists.

▸ **Turn on push protection first — it's the cheapest breach you'll never have.**

Demo 3 · Copilot builds it, GitHub keeps it safe

TD SYNNEX

LIVE DEMO · 15 MIN Just watch

1 Rules and a branch

New branch from the status bar. Copilot reads our copilot-instructions.md; switch Chat to Agent.

2 Build it

Prompt 1: a GitHub cost calculator from prices.json. Review, keep, run: $6,940 a month for 100 developers.

3 The shortcut

Prompt 2: fill the developer count from our org. Paste a token into .env, commit, Publish Branch.

4 Blocked

Push protection stops the push and names the secret. Nothing reached GitHub.

5 Fix it and ship it

Undo, ignore and delete .env, keep the token in a Codespaces secret. Pull request, green check, merge.

WATCH FOR ✓ **Copilot's changes before you keep them** ✓ **the secret named in the block** ✓ **no token in the code**

IF IT BREAKS if Copilot goes sideways, discard and merge in answer-key. If the push isn't blocked, delete the token now.

TD SYNNEX

06

Admin essentials & buying

PART 6 — What it costs, and how partners buy · 6 min

What it costs: per user, per committer, per credit

TD SYNNEX

GITHUB ENTERPRISE

$21

per user / month *

Cloud or Server, same license. Org members and outside collaborators on private or internal repos consume a seat; billing managers and guests do not. 30-day trial, up to 50 seats.

COPILOT

$19 · $39

per user / month

Business $19 with 1,900 AI credits; Enterprise $39 with 3,900 (needs Enterprise Cloud). Credits pool; completions are unmetered; overage $0.01 per credit unless capped.

SECURITY

$19 · $30

per active committer / month

Secret Protection $19; Code Security $30. A committer is counted once across all repos, 90-day window. Free on public repos; bundle = Advanced Security.

CONSUMPTION

PAYG

metered monthly

Actions minutes beyond 50,000 (per minute by runner size and OS) · Codespaces from $0.18 per hour (2-core) plus $0.07 per GB-month · Packages beyond 50 GB · Git LFS.

WORKED EXAMPLE

100 developers, 60 of them committing code: 100 × $21 Enterprise + 100 × $19 Copilot Business + 60 × ($19 + $30) security = **$6,940 per month** before Actions, Codespaces and storage.

* github.com/pricing (Aug 2026) lists Enterprise at $21 and Team at $4 "per user/month for the first 12 months" — renewal pricing is not published; confirm it before quoting. Copilot: self-serve (card/PayPal) Business and Enterprise seats are paid before access from 1 Sep 2026.

- **Three meters — seats, committers, credits — size all three before you quote.**

Partners buy it through Azure — and can run it for the customer

TD SYNNEX



1 Azure subscription

The customer's own Azure subscription — under their EA, or the Azure plan you provide as a CSP. Free-trial, sponsorship and legacy pay-as-you-go offers are not accepted.

2 Metered billing

GitHub Enterprise, Copilot, Secret Protection, Code Security, Actions and Codespaces usage all land on the Azure invoice.

3 Create the enterprise

As a CSP partner: Azure portal → GitHub Enterprise → create the customer's enterprise, choose managed or personal accounts, link the subscription.

4 Hand over, stay attached

Invite the customer as enterprise owner; downgrade yourself to billing manager — no seat consumed, renewals stay in your hands.

ALSO

Direct from GitHub

card or invoice, yearly volume/subscription — the classic route

Existing agreements

prepaid or volume deals keep billing their way until they expire

Trial first

30 days, up to 50 seats, Advanced Security included; Copilot is not

▸ **The Azure subscription is the purchase order — everything GitHub sells can ride on it.**

Take it home: run the demos yourself

TD SYNNEX

DEMO 1

Version history you control

1. Create a repo on github.com
2. Clone it in VS Code
3. Save a version after each change
4. Bring a deleted slide back

YOU NEED
VS Code, git, and PowerPoint

DEMO 2

A dev environment in one click

1. Copy the starter template
2. Open it in a codespace
3. Look at the one-file setup
4. Stop it when you're done

YOU NEED
a browser and a free account

DEMO 3

Copilot builds it, safely

1. Paste the prompts into Copilot
2. Run the calculator
3. Push a test token, see the block
4. Move it to a Codespaces secret

YOU NEED
Copilot Free and a public repo

EVERYTHING IS ONLINE Guides, starter files, slides: **github.com/TD-Synnex-Cloud-Engineering/Microsoft-Technical-Enablement**

▸ **An hour on your own this week is what makes today stick.**

Making IT Personal™

Where to go next

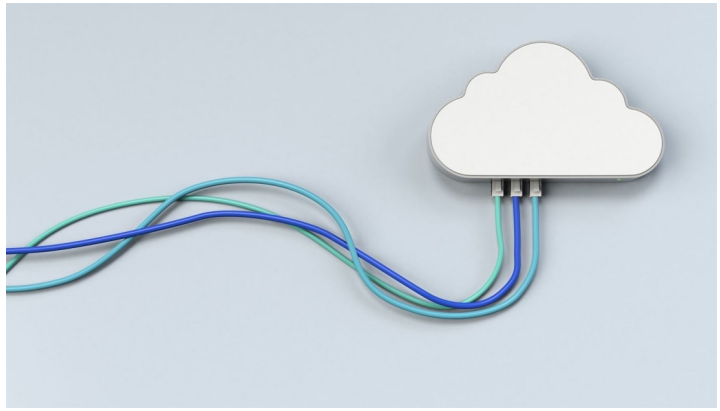
TD SYNNEX

- **Run the demos yourself** — the take-home slide has the steps, and everything works on a free account.
- **GitHub Skills** — free hands-on courses inside GitHub — "Introduction to GitHub" and "Review pull requests" are the next two.
- **GitHub Foundations certification** — the entry-level exam that maps almost one-to-one onto today.
- **Microsoft Learn** — the free GitHub Foundations and GitHub Administration learning paths.
- **Bring us a customer** — book a 1:1 with Cloud Enablement — we will scope identity, policies and licensing with you.

Making IT Personal™

Cloud Enablement Services

TD SYNNEX



CloudEnablementServices@TDSynnex.com

To view and register, sign in to TD SYNNEX's LMS site — create a free account if you don't have one.

Ask me for direct links to any class below.

Azure Virtual Desktop (AVD)	1:1 workshop (2 days) · walkthrough (5–6 hr) · video series
Azure Migrate	1:1 workshop (4 hr) · walkthrough (4 hr) · video series
Microsoft Power Apps	1:1 workshop (3 hr) · walkthrough (2–3 hr) · video series
Windows 365 & Intune	walkthrough (4–5 hr) · video series
Microsoft 365 Security	walkthrough (4–5 hr) · video series
Microsoft Cost Management	walkthrough (2 hr)
Azure Log Files	walkthrough (2 hr)
Microsoft Power BI	walkthrough (2–3 hr)
Azure Security	walkthrough (2 hr)
Azure Arc	walkthrough (1–2 hr)
Microsoft Sentinel	walkthrough (3 hr)
Microsoft Fabric	walkthrough (2 hr)
GitHub	walkthrough (2 hr) — this class

1:1 = hands-on with an instructor · walkthrough = 1-to-many demo class · video series = self-paced on the LMS

Making IT Personal™

TD SYNNEX

THANK YOU

Russ Seaman

Solutions Architect II · TD SYNNEX Cloud Enablement

russ.seaman@tdsynnex.com

linkedin.com/in/russseaman

These slides and links
github.com/TD-Synnex-Cloud-Engineering/Microsoft-Technical-Enablement



Making IT Personal™